17.4 Security technologies in E-Commerce transaction

Since a large amount of confidential information are involved in E-Commerce activities it must be transmitted through safe and secured network. Sophisticated security technologies are required to ensure the security of E-Commerce transactions. At present, the security technologies in E-Commerce transactions are roughly classified into

- Encryption technology
- Authentication technology
- Authentication protocols

17.4.1 Encryption technology

Encryption technology is an effective information security protection. It is defined as converting a Plaintext into meaningless Ciphertext using encryption algorithm thus ensuring the confidentiality of the data. The encryption or decryption process uses a key to encrypt or decrypt the data. At present, two encryption technologies are widely used. They are symmetric key encryption system and an asymmetric key encryption system.

Symmetric key encryption

The Data Encryption Standard (DES) is a Symmetric key data encryption method. It was introduced in America in the year 1976, by Federal Information Processing Standard (FIPS).

DES is the typical block algorithm that takes a string of bits of cleartext (plaintext) with a fixed length and, through a series of complicated operations, transforms it into another encrypted text of the same length. DES also uses a key to customize the transformation, so that, in theory, the algorithm can only be deciphered by people who know the exact key that has been used for encryption. The DES key is apparently 64 bits, but in fact the algorithm uses only 56. The other eight bits are only used to verify the parity and then it is discarded.

Today, it is considered that DES is not safe for many applications, mainly because of its relatively smaller key size (56-bit).

Asymmetric or Public key encryption

Asymmetric encryption also called as RSA (Rivest-Shamir-Adleman) algorithm.

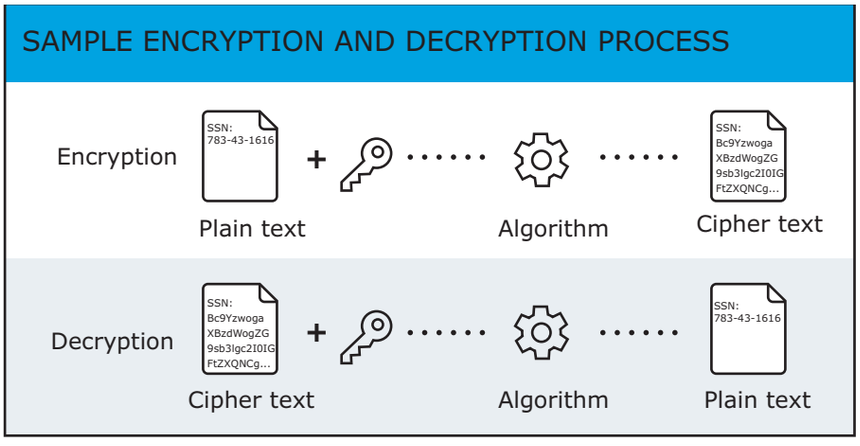


Figure 17.1 Data Encryption and Decryption process

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